SFDN: A Novel Semantic Feature Decouple Network for Fine-Grained Remote Sensing Object Detection

Xi Yang, *Senior Member, IEEE*, Zhongyuan Zhou, and Dong Yang

***Abstract*—Fine-grained object detection (FGOD) aims to identify subcategories of objects by extracting more discriminative semantic information. Bounding box regression typically requires detailed texture and edge information to accurately delineate object boundaries, while classification requires richer semantic information. However, existing methods use the same input features in the model, resulting in an imbalance between the localization task and the fine-grained classification task. To address this issue, we propose a novel semantic feature decouple network (SFDN) that effectively separates semantic information for localization and fine-grained classification. For the localization task, we propose a regression feature fusion module (RFFM) to extract feature maps with more edge information. To enhance classification one, we propose a fine-grained feature diversification module (FFDM) to capture discriminative semantic information from feature maps by introducing fake attention maps. Aiming to extract richer fine-grained semantic information, we propose an adaptive local perception module (ALPM) to deeply extract multiscale semantic feature information by using dilated convolution at varying dilation rates. Extensive experiments demonstrate that the proposed network, respectively, achieves the mean average precision (mAP/%) of 80.08% and 49.44% on the ShipRSImageNet (SRSIN) and FAIR1M-v1.0 datasets, outperforming state-of-the-art (SOTA) methods by 3.95% and 1.46%.**

***Index Terms*—Feature decoupling, fine-grained object detection (FGOD), semantic information.**

I. Introduction

REMOTE sensing object detection, as a subfield of computer vision, focuses on the identification and localization of objects of interest in aerial images, such as vehicles or aircraft [1]. With the increased clarity of remote sensing images (RSIs), there is a concurrent increase in the semantic information, which necessitates the development of more efficient interpretation methods. In recent years, the rapid evolution of deep learning has significantly propelled the progress of object detection technology [2], [3], [4], [5], [6], [7], [8]. Consequently, with the introduction of a series of high-efficiency detectors that achieve high-precision results, significant improvements have been made in object detection for RSIs [9], [10], [11], [12], [13], [14], [15]. However, fine-grained object detection (FGOD) within RSIs poses a formidable challenge. Since the textural features among various subcategories are similar to a certain degree, discriminative semantic information is required to identify fine-grained subcategories.

High-definition RSIs are rich in semantic information, and the classification and localization tasks of object detection heavily rely on the semantic information encoded by advanced semantic-rich features [16], [17]. To improve the detection of objects at various scales, the feature pyramid network (FPN) [18] effectively distills the semantic information from high-level semantic-rich features and integrates it into low-level feature maps that preserve fine details. This architectural design enhances the availability of semantic information for early-stage features and facilitates the detection of objects at various scales. Actually, bounding box regression typically requires detailed texture information and edge cues to accurately delineate object boundaries, while object classification benefits from richer semantic information [19], [20]. Previous methods [3], [4], [21] have commonly addressed this issue by employing a shared head network for both tasks on each level of the feature maps, tailored to a specific object scale. The semantic misalignment between the two tasks has been recognized and noted by subsequent works such as Double-Head R-CNN [22] and TSD [23]. While these decoupled-head approaches represent a significant advancement by employing separate network branches with distinct parameters for classification and localization, they share a fundamental limitation, which means that both branches operate on the same set of input features. This design only mitigates but does not fundamentally resolve the feature conflict, as the shared input features inherently carry the biases and preferences of the backbone network, which may be optimized for one task at the expense of the other. Consequently, the decoupled heads struggle to learn truly task-specific feature representations, ultimately limiting their effectiveness in FGOD where the demands for discriminative semantics and precise localization are exceptionally high.

Nevertheless, the fine-grained classification and localization tasks in RSIs pose new challenges. For instance, the DOTA [24] dataset, one of the largest remote sensing datasets with oriented bounding box (OBB) annotations, comprises

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Xi Yang is with Xidian University, Xi'an 710071, China (e-mail: yangx@xidian.edu.cn).

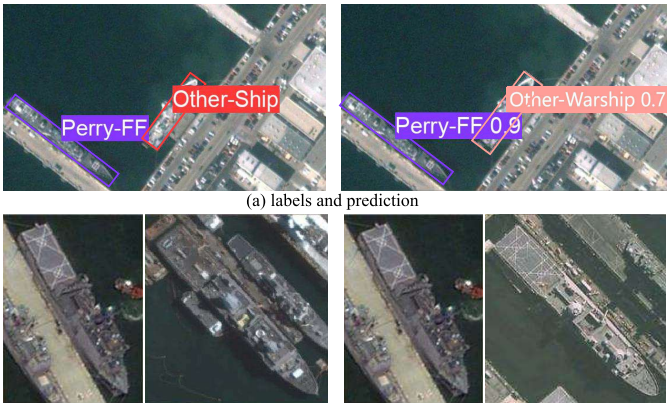
Zhongyuan Zhou is with Hangzhou Institute of Technology, Xidian University, Hangzhou 311231, China (e-mail: xdzzy@stu.xidian.edu.cn).

Dong Yang is with Xi'an Institute of Space Radio Technology, Xi'an 710100, China (e-mail: yangd@cast504.com).

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(a) labels and prediction

(b) intra-class differences and inter-class similarities

Fig. 1. Difficulties and challenges in fine-grained remote sensing object detection. (a) Difficulty in fine-grained classification. Other-ship and other-warship are object categories that are not annotated in detail. It is a difficult sample for FGOD. (b) Images on the left represent the difference in performance of the same subcategory, while the images on the right represent the similarity in performance of different subcategories.

15 categories. In contrast, the FAIR1M-v1.0 [25] dataset, the latest remote sensing dataset for fine-grained tasks, consists of five major categories and 37 subcategories. The appearance of these categories within the DOTA dataset exhibits significant variations, and even in RSIs, differentiating between distinct categories, such as bridges and aircraft, is not excessively difficult. However, most existing methods demonstrate satisfactory performance on the DOTA [24] dataset but struggle to accurately identify fine-grained objects within the FAIR1M-v1.0 [25] dataset.

In practice, the classification and localization tasks have different objectives. The classification task focuses on extracting effective semantic information to assign appropriate class labels to objects, while the localization task focuses on obtaining detailed boundary information to accurately locate objects. As shown in Fig. 1(a), networks struggle to distinguish highly similar objects, thereby necessitating the enhancement of discriminative feature extraction capabilities.

As shown in Fig. 1(b), compared to natural images, RSIs in fine-grained classification lack multiangle feature information, and the presence of many tiny objects makes appearance-based identification difficult. Actually, environmental differences surrounding objects in RSIs can provide valuable clues about their shape, orientation, and other features. Expanding the receptive field to increase the extraction of environmental information assists in extracting these valuable clues about objects' shape, orientation, and other features, thereby enhancing the network's ability to extract discriminative features. The effectiveness of expanding the receptive field has been validated in Transformer [26], highlighting its importance in achieving successful outcomes [27]. Additionally, designing convolutional networks with large receptive fields or specific structures can also compete effectively with Transformer-based models. For example, ConvNeXt [28] utilizes deep convolutions to significantly improve downstream task performance, while RepLKNet [29] achieves impressive results by reparameterizing with large convolutional kernels.

To tackle these challenges, we propose a novel method where features are disentangled into localization features and fine-grained classification features, denoted as the semantic feature decouple network (SFDN). Unlike mainstream decoupling methods [19], [22], [23] that merely separate the processing heads, our SFDN performs decoupling at the feature level, generating dedicated input features for each task. Our SFDN consists of fine-grained feature diversification module (FFDM), regression feature fusion module (RFFM), and adaptive local perception module (ALPM). The proposed SFDN demonstrably enhances the efficacy of fine-grained detection.

For the fine-grained classification task that demands discriminative semantic features, we propose the FFDM to effectively extract fine-grained semantic information by introducing fake attention maps. Current object detection methods use coarse-grained classification based on salient features, which is insufficient for fine-grained subcategories due to intercategory similarities and intracategory differences. FFDM combines feature maps from different pyramid levels to create a semantically rich feature map, thereby enhancing classification accuracy. It generates multihead attention maps to weight these features, producing fine-grained features. To improve attention quality, we introduce fake attention maps, compelling the model to focus on both salient and suboptimal features. The final prediction is derived by comparing the real and fake attention map predictions.

For the localization task that requires more detailed information on object boundaries, we propose the RFFM to extract high-resolution feature maps enriched with additional edge information. RFFM combines feature maps from adjacent levels, merging detailed edge features with comprehensive object views to generate a final feature map, thereby improving localization accuracy efficiently.

In order to extract richer semantic information, we propose the ALPM to obtain the semantic information of different receptive fields. ALPM also tackles FGOD challenges including the loss of multiangle feature information in RSIs and the fading of tiny object features during forward propagation due to repeated pooling and downsampling. Specifically, ALPM applies dilated convolutions with rates 1, 3, and 5 on feature map C5 to capture multiscale context without sacrificing spatial resolution. We gather context information from different receptive fields, which involves convolution, concatenation, and softmax operations to create spatial adaptive weights and aggregate context information through weighted sums. By integrating these enhanced features top-down into the FPN, it enriches semantic information for classification and improves precise multiscale localization. Its design captures rotational equivariance via contextual relationships, directly addressing the orientation and scale variability illustrated in Fig. 1(b), while its multiscale approach preserves features of tiny objects typically lost in standard architectures.

The main contributions of this article are given as follows.

1) For the fine-grained classification task, we propose the FFDM to effectively extract discriminative fine-grained semantic information by introducing fake attention maps. FFDM facilitates the learning of attention and

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fake attention feature maps, enhancing the quality of attention for fine-grained features.

2) For the localization task, we propose the RFFM to extract high-resolution feature maps enriched with additional edge information. RFFM extracts texture details and boundary information, which is beneficial for accurate localization tasks.
3) We propose the ALPM, designed to deeply extract multiscale semantic feature information. ALPM contributes to a more comprehensive understanding of the contextual intricacies involved in FGOD.

II. Related Work

A. Object Detection in RSIs

Current remote sensing object detectors are broadly categorized into two-stage [4], [21] and one-stage [2], [3] frameworks. The two-stage RoI transformer [9] utilizes fully connected layers to rotate candidate horizontal bounding boxes (HBBs) in the first stage. Subsequently, features within these rotated bounding boxes undergo extraction for further regression and classification. SCRDet [30] employs an attention mechanism to mitigate background noise, enhancing the modeling of crowded and small objects. FFEDet [31], specifically designed for small object detection in RSIs, also employs the attention mechanism to emphasize relevant features in adjacent scales, which effectively solves the challenges brought by complex background and interference feature expression. Addressing the instability of training losses stemming from rotation angle periodicity, Oriented RCNN [32] and Gliding Vertex [33] introduce novel box encoding systems. SASS [34] proposes a pretraining paradigm with simple adversarial training or even no adversarial training. Certain approaches [35], [36], [37], treating remote sensing detection as a point detection task [38], offer an alternative perspective to resolve challenges in this domain.

Different from the two-stage detection frameworks, the one-stage ones directly classify and regress OBBs from densely sampled anchors on a grid. The one-stage S2A network [10] achieves robust object feature extraction through oriented feature alignment and orientation-invariant feature extraction. In contrast, DRN [39] leverages attention mechanisms to dynamically refine the features extracted from the backbone, leading to more accurate predictions. Unlike Oriented RCNN and Gliding Vertex, RSDet [40] addresses the discontinuity of regression loss by introducing a modulated loss. AOPG [41] and R3Det [42] adopt a progressive regression approach, refining bounding boxes from coarse-to-fine granularity. REPAN [43] explores semantic associations and extracts distinguishing features among multiple parts to tackle the complex weak association and fuzzy boundary problems. Angle-based regression, as adopted by some works [44], [45] based on rotational region CNN (R2CNN) [45], introduces an angle parameter to the regression branch. In contrast, point-based regression, applied in these works [33], [40], directly predicts offsets or coordinates of the four vertices of OBBs. Beyond traditional CNN-based frameworks, DETR [46], into the realm of remote sensing detection tasks, thereby contributes to increased research diversity.

B. Fine-Grained Object Detection

In contrast to general object detection, FGOD requires the precise positioning of objects and the accurate classification of objects with high appearance similarity. Zhang et al. [47] design an additional backbone to extract attribute information, such as aspect ratio and scale, and utilize SVM to make the final classification based on concerted attribute features and multiscale image features. Sun et al. [48] employ a cascade R-CNN structure with different NMS thresholds in each stage to realize coarse-to-fine recognition. These strategies prolong the recognition process and increase the computational burden. Fang et al. [49] find that the detection performance for objects with a high aspect ratio is much lower than the overall average. The mainstream strategy to deal with the excessive aspect ratio of ships is still increasing the range of aspect ratios of anchors. The cluster-based approach proposed in [49] is an effective attempt, although it is not end to end. To improve the detection performance of complex scenes such as high-similarity objects and dense distributions, feature fusion and attention mechanisms are widely used and developed in FGOD to enhance the feature representation capabilities of the network. For example, [44] and [50] employ dilated convolution and deformable convolution, respectively, to strengthen multiscale semantic information in the fusion process. There are also some new networks that fuse features by constructing new backbones. Dual-tunnel CNN [51] analyzes the joint features from the fused information spatially and spectrally. Zhang et al. [52] design a PToP CNN to cover a convolutional filter bank with three tunnels. Unfortunately, most feature fusion structures only consider either top-down or bottom-up fusion direction, whereas another information flow is ignored. A novel feature fusion called MSR-FPN [53] achieves high performance by merging the top-down and bottom-up branches through a dilated convolutional parallel structure, while LR-FPN [54] achieves it in RSIs by enhancing shallow refined location information extraction and fine-grained context interaction.

The adoption of a decoupled head has long been the standard configuration for one-stage detectors [55], [56]. Recent research, exemplified by Double-Head R-CNN [22] and TSD [23], revisits the specialized sibling head widely used in the R-CNN family [4], [21], [57], [58], finally discerning the essence of the task misalignment between classification and localization. YOLOX [59] underscores that the coupled detection head may compromise performance, introducing the decoupled head to the YOLO family [3], [60], [61] for the first time. This innovation significantly improves convergence speed and boosts overall performance. Building on the decoupled head, DDOD [19] proposes using deformable convolutions to learn separate convolutional offsets for each branch, aiming to adaptively select specific spatial features for each head. These works collectively emphasize the importance of decoupling between classification and localization tasks. However, these mainstream decoupling methods primarily operate at the head level, employing separate convolutional layers or branches for classification and regression. While effective in learning task-specific feature contexts, they share a critical limitation: both branches receive identical input features from the backbone

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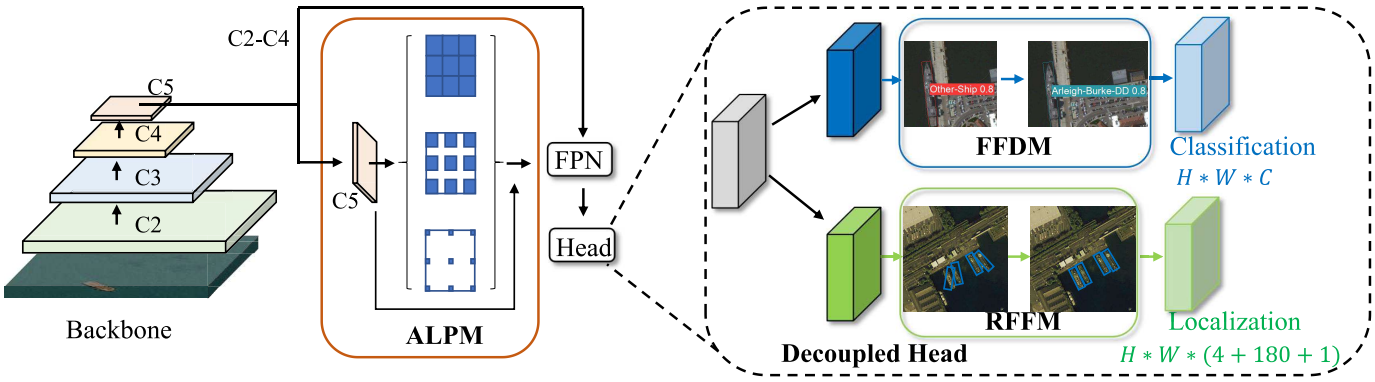


Fig. 2. Architecture of our SFDN. FFDM facilitates the learning of attention and fake attention feature maps, enhancing the quality of attention for fine-grained features. RFFM extracts texture details and boundary information, which is beneficial for accurate localization tasks. ALPM deeply extracts multiscale semantic feature information.

and FPN. These shared features are inherently optimized for a general purpose, representing a compromise between the conflicting demands of precise localization and robust classification. Consequently, even with dedicated processing heads, the performance of each task remains constrained by the suboptimal shared input features. The decoupled heads are forced to "undo" the inherent biases in the features before learning task-specific representations, which limits the ultimate effectiveness of the decoupling. For example, DDOD [19] attempts to address this by using deformable convolutions to provide more flexible receptive fields for each task, yet the features adjusted by these convolutions still originate from the same shared feature maps, thus not fully overcoming the fundamental constraint. Inspired by these advancements, TSCODE [62] proposes decoupling semantic features at the head, effectively resolving the conflict between positioning tasks and classification tasks to a certain extent. However, it is crucial to note that these efforts primarily alleviate the conflict between localization and classification, and fine-grained classification imposes higher requirements for discriminative feature semantics. This limitation becomes particularly pronounced in FGOD, where the subtleties distinguishing subcategories are often faint and can be easily corrupted or lost in features that are generically optimized for both tasks.

In addition, SIM-Trans [63] integrates object structural information from images through its SIL and MFB modules, thereby enhancing the transformer model's learning of discriminative representations and ultimately improving accuracy in fine-grained visual classification. CF-ORNet [64] integrates an additional backbone to learn fine-grained classification while maintaining lightness through knowledge distillation. By leveraging prototype contrastive learning, PCLDet [65] maximizes interclass distances while minimizing intraclass distances. To better capture discriminative features, SFR-Net [66] introduces spatial and channel transformers and employs metric learning to enhance the separability of fine-grained classes. RB-FPN [67] refines features extracted by FPN using attention modules and label allocation strategies, dynamically modifying the IoU threshold based on statistical features. FEFP-Net [68] is proposed to effectively alleviate the challenges of interclass feature similarity, object size variability, and inconsistency between location and classification predictions based on multilevel fine-grained features. Panoptic perception [69] model, based on joint optimization, integrates pixel-, instance-, and image-level information that is captured from coarse to fine-grained. AMFA [70] uses two attention mechanisms to identify distinguishing regions and local features corresponding to subtle visual features. PETDet [71] optimizes the fine-grained recognition task in the two-stage FGOD by improving proposal generation and feature extraction. DRNet [72] generates discriminative representations by re-encoding object features through a dual refinement mechanism that introduces fine-grained branches working in parallel. It further enhances representational discrimination by designing strategies to minimize confusion loss and employ cross-validation. QEDetr [73] proposes a baseline detector by combining denoising training and angle encoding, and introduces a novel attention mechanism and multiscale fusion method to extract fine-grained features.

Different from the above decoupling methods in FGOD, our SFDN represents a paradigm shift from head-level to feature-level decoupling. Instead of processing shared features with different heads, we generate dedicated input features specifically engineered for each task's requirements. This approach moves beyond the limitations of previous decoupling strategies by addressing the problem at the feature generation stage rather than attempting to compensate for it in the processing stage. By providing the localization branch with high-resolution, edge-enriched features via RFFM and the classification branch with discriminative, semantics-enhanced features via FFDM, we enable each task to operate on features that are inherently aligned with its objectives, thus achieving a more complete and effective decoupling.

III. Methods

The overall architecture of the SFDN is shown in Fig. 2. The proposed method adheres to a standard one-stage detection framework, comprising the backbone, the feature pyramid, and the detection head. The backbone and feature pyramid collaboratively generate multiscale feature maps based on the input image. Subsequently, our proposed ALPM is employed to extract multiscale semantic feature information at the final layer of the backbone. Finally, the semantic decoupled head processes three levels of feature maps and produces decoupled

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feature maps that are specifically tailored for fine-grained classification and localization tasks.

Fine-grained classification and localization represent two interconnected yet inherently conflicting objectives within object detection. Achieving fine-grained classification necessitates richer semantic information for each object, whereas localization demands more intricate boundary information. The prevalent design approach involves decoupled heads, wherein the feature projection functions for classification and localization share a common structure but are learned with distinct parameters to impart different feature contexts to each task. However, this conventional design falls short of fully addressing the issue, as the semantic information is still significantly influenced by the shared input features. While recent work [19] endeavors to enhance feature learning with more flexible contexts for each task using deformable convolutions, their effectiveness is constrained by the fact that these features still originate from the same shared input, limiting their capacity to fully resolve the conflict between fine-grained classification and precise localization.

To address this issue, the proposed SFDN separates the feature encoding for the FGOD task into distinct branches: a fine-grained classification branch and a localization branch. Within these branches, diverse semantic feature mappings are employed, contributing to an enhanced overall performance. This decoupling strategy allows for more specialized treatment of each aspect of the task, ensuring that the unique requirements of fine-grained classification and precise localization are addressed effectively. Given a ground-truth object assigned to a specific pyramid level i, with bounding box y_b and class label y_c. The total loss $\mathcal{L}$ is given as follows:

$$\mathcal{L} = \mathcal{L}_{\text{cls}}\left(f_c\left(x_i^{\text{cls}}\right), y_c\right) + \lambda \mathcal{L}_{\text{loc}}\left(f_l\left(x_i^{\text{loc}}\right), y_b\right) \tag{1}$$

where $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{loc}}$ are the loss of classification and localization. $f_c(\cdot)$ and $f_l(\cdot)$ are classification and localization branches, while x_i^{cls} and x_i^{loc} are the decoupled input features of them.

The classification task is one of the points of the FGOD task. The fine-grained features need to be captured to achieve accurate discrimination for similar classes. Therefore, we generate more discriminative semantic feature maps. The aim of the localization is to attain high recall and precision in bounding box regression. To achieve this goal, we furnish this branch with feature maps that encapsulate more detailed texture and boundary information. This method ensures that the localization branch is equipped with comprehensive and nuanced spatial details, enhancing the accuracy of bounding box regression.

To capture the semantics for facing the challenges in Fig. 1, FFDM first introduces random fake attention maps that encourage the model to focus on both salient and nonsalient features, thereby learning discriminative semantic features by attending to both clear and subtle cues in the image. The fake attention mechanism forces the model to diversify its attention across different aspects of the object, improving its ability to capture the fine-grained distinctions needed for accurate classification. Then, RFFM fuses edge information from lower pyramid levels with semantic context from higher pyramid levels. It also incorporates features from higher pyramid levels, which capture global semantic information such as object category and context. Finally, ALPM utilizes dilated convolutions with varying dilation rates to capture multiscale semantic features while maintaining spatial resolution. It adaptively fuses these multiscale features to provide a more comprehensive view of the context around the object. Here, ALPM's multiscale dilation captures rotational equivariance through contextual relationships for the challenge of multiangle feature loss in Fig. 1(b).

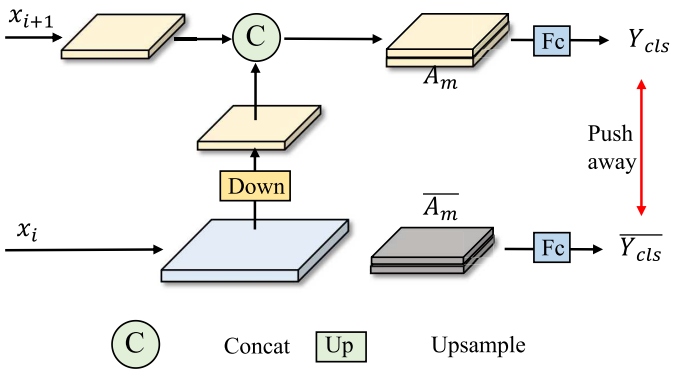


Fig. 3. Architecture of FFDM.

Our SFDN represents a significant breakthrough beyond existing decoupling strategies. While previous methods [19], [22], [23] focus on learning different feature contexts from the same input, our approach provides each task with features that are optimized from the ground up for its specific objective. The localization branch receives features enriched with edge and texture details through RFFM, while the classification branch receives features enhanced with discriminative semantic information through FFDM. This eliminates the inherent compromise present in shared features and allows each branch to achieve its maximum potential without interference.

A. Fine-Grained Feature Diversification Module

The prevailing classification tasks in current object detection methods tend to adopt a coarse-grained method, relying on the characteristics of key or prominent parts of the object, commonly known as salient features. However, when dealing with fine-grained subcategories, there exist both intercategory similarities and intracategory differences. Different subcategories share similarities due to comparable backgrounds and morphologies, while the same subcategories exhibit differences due to varying backgrounds and morphologies. Relying solely on salient features is inadequate for achieving accurate fine-grained classification. Therefore, there is a compelling need to extract more discriminative semantic features to address the nuances of fine-grained classification accurately. In general, in this section, we address the core challenge in fine-grained classification, where objects within the same class often exhibit significant variations in appearance, while objects across classes may share similar characteristics such as small changes in texture or orientation. This challenge is exacerbated in RSIs where objects of interest, such as ships, planes, or vehicles, may appear in diverse contexts and at different scales.

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Therefore, we propose an FFDM for efficient and accurate fine-grained classification. Specifically, at each pyramid level i, FFDM leverages the feature maps from two levels, x_i and x_{i+1}, to generate a semantically rich feature map for classification. As shown in Fig. 3, we downsample x_i by a factor of 2 and concatenate it with x_{i+1} to generate the final x^{cls}

$$x_i^{\text{cls}} = \text{Down}\left(x_i\right) \oplus x_{i+1} \tag{2}$$

where $\oplus$ and Down$(\cdot)$ represent concatenation and a downsampling convolutional layer. This structure can encode more semantic information.

The resulting feature maps x^{cls} are denoted as $x^{\text{cls}} \in \mathbb{R}^{C*H*W}$, where H, W, and C represent height, width, and the number of channels, respectively. Subsequently, M multihead attention maps are generated by applying a simple convolutional function to the extracted features x^{cls}. These attention maps are represented as $A = \phi(x^{\text{cls}})$, where $\phi(\cdot)$ signifies the convolutional function with a kernel size of 1×1 and the number of attention maps M. The set of attention maps generated is denoted as $A = \{A_1, A_2, \ldots, A_M\}$. Notably, $A_M \in A$ represents the attention maps that highlight interesting regions in the input image. The feature maps x^{cls} are then weighted by the attention maps to generate fine-grained features G_m after global average pooling

$$G_m = \phi\left(x^{\text{cls}} \cdot A_m\right) \tag{3}$$

where ϕ represents the global average pooling operation, and $\cdot$ denotes elementwise multiplication. The final prediction $Y_{x^{\text{cls}}}(A)$ is obtained by applying a classifier to the sum of the weighted features

$$Y_{x^{\text{cls}}}\left(A\right) = fc\left(\sum_{m=1}^{M} G_m\right) = fc\left(\sum_{m=1}^{M} \phi\left(x^{\text{cls}} \cdot A_m\right)\right) \tag{4}$$

where fc$(\cdot)$ represents a classifier including a fully convolutional layer and a batch normalization layer.

To enhance attention quality for fine-grained features, we introduce the concept of fake attention maps. The fake attention map is an attention map simply generated by a random function. The random generation is inspired by the need for the model to be robust and to avoid overfitting to specific patterns. The random generation strategy for fake attention maps is grounded in information entropy theory. By maximizing entropy in attention distribution, we force the model to explore a wider range of features within nonsalient regions, avoiding over-reliance on dominant features. The sparsity constraint ensures limited overlap with object regions. Noting that a random sampling function for fake attention maps avoids bias, improves generalization, reduces overfitting, and ensures the model focuses on subtle, discriminative features across diverse regions of the image. This flexibility and lack of structure give the model the freedom to learn robust and nonbiased features for fine-grained classification, unlike more rigid strategies such as uniform sampling, Gaussian sampling, random Gaussian noise, or spatially sparsified sampling, which could inadvertently limit the model's learning capacity. Therefore, the fake prediction $Y_{x^{\text{cls}}}(\overline{A})$ is expressed as follows:

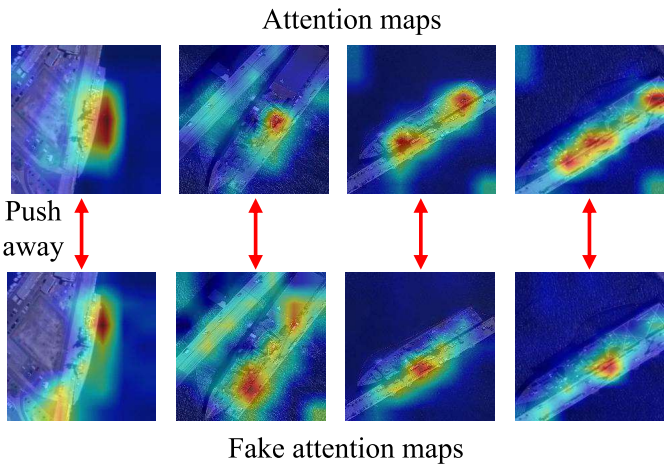


Fig. 4. Relationship between the attention map and the fake attention map.

$$Y_{x^{\text{cls}}}\left(\overline{A}\right) = fc\left(\sum_{m=1}^{M} \overline{G_m}\right) = fc\left(\sum_{m=1}^{M} \phi\left(x^{\text{cls}} \cdot \overline{A_m}\right)\right) \tag{5}$$

where $\overline{A}$ represents fake attention maps. The randomly generated fake attention maps are designed to be independent of object positions, avoiding bias toward object centers or predefined areas. By introducing these maps, the model explores diverse feature areas, helping to avoid overfitting and improve generalization. This process encourages the network to focus on generalized, nonsalient features, making it more robust to variations in object appearance. The sparse, random distribution of fake attention maps ensures they do not excessively overlap with the objects, promoting a more balanced feature learning process.

As shown in Fig. 4, FFDM facilitates the learning of attention and fake attention feature maps, enhancing the quality of attention for fine-grained features. The fake attention maps in FFDM are randomly generated during training to diversify the network's focus, compelling it to learn broader feature representations by balancing attention between salient and nonsalient regions. Simultaneously, learning fake attention feature maps reduces attention to salient features, suppressing background attention proportionately. Specifically, we compel the attention algorithms to focus on both salient and suboptimal features, comparing the attention maps with fake attention maps. It also forces the model to incorporate less obvious yet discriminative details, enhancing its ability to distinguish fine-grained subcategories with high visual similarity. Different from the contrastive learning based on data augmentation, our fake attention maps are applied to the network during training without the need for manually defining positive or negative pairs, reducing reliance on predefined pairs and making the model more flexible and effective in capturing subtle features for fine-grained classification tasks.

The final prediction Y_{cls} is obtained by taking the difference between the attention prediction $Y_{x^{\text{cls}}}(A)$ and its fake prediction

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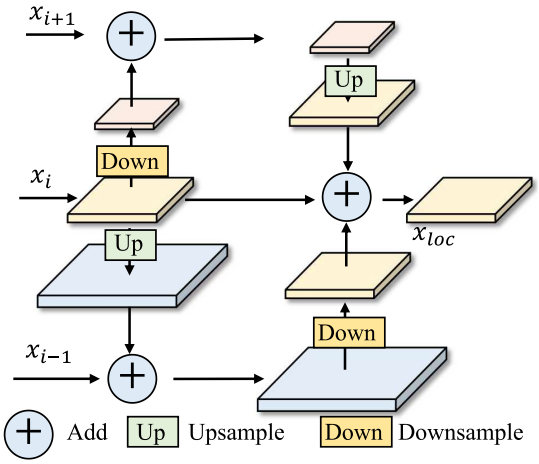


Fig. 5. Architecture of RFFM.

$Y_{x^{\text{cls}}}(\overline{A})$

$$Y_{\text{cls}} = Y_{x^{\text{cls}}}(A) - Y_{x^{\text{cls}}}\left(\overline{A}\right). \tag{6}$$

Finally, FFDM enhances the classification branch by guiding the network to focus on discriminative features for fine-grained object identification. The output of FFDM is then passed into the classification head, where a final decision is made based on the weighted features.

B. Regression Feature Fusion Module

In contrast to classification, localization is a task that necessitates intricate texture details and boundary information to accurately predict the angles of an object. The feature maps in lower pyramid levels exhibit a more robust response to contours, edges, and detailed textures of objects.

Traditionally, features at different levels carry positional information about objects of various sizes. Larger features encompass low-dimensional texture details and positions of smaller objects. In contrast, smaller features contain high-dimensional information and positions of larger objects. FPN [18] provides an efficient architectural design for fusing multi-scale features through cross-scale connections and information exchange, thereby boosting the detection accuracy of objects of varied sizes.

In order to combine the high-dimensional information of the deep feature map with the detailed information of the shallow feature map, we propose the RFFM to extract scale-invariant features and achieve precise localization without compromising efficiency. At each pyramid level i, our RFFM takes feature maps from levels x_{i-1}, x_i, and x_{i+1}. x_{i-1} contributes more detailed and edge features, while x_{i+1} provides a more comprehensive perspective of objects. The structure of RFFM is depicted in Fig. 5. Initially, x_i is upsampled by a factor of 2 and aggregated with x_{i-1}. It preserves high-frequency edge information from x_{i-1} through skip connections, reducing boundary blurring. Subsequently, a 3×3 convolutional layer with a stride of 2 downsamples to match the resolution of x_i. Following this, x_i is downsampled, aggregated with x_{i+1}, and then upsampled and aggregated to generate the final x^{loc}. The computation is expressed as follows:

$$\begin{aligned} x_i^{\text{loc}} = x_i &+ \text{Up}\left(x_{i+1} \oplus \text{Down}\left(x_i\right)\right) \\ &+ \text{Down}\left(x_{i-1} \oplus \text{Up}\left(x_i\right)\right) \end{aligned} \tag{7}$$

where Up(·) and Down(·), respectively, represent upsampling and downsampling convolutional layer. Here, we employ bilinear interpolation with a scale factor of 2 for Up(·), followed by a 3×3 convolutional layer with stride 1 and padding 1. This convolutional layer uses ReLU activation and maintains the same number of channels as the input feature map. The bilinear interpolation ensures smooth feature expansion, while the convolutional layer helps to refine the upsampled features. For Down(·), we utilize a 3×3 convolutional layer with a stride of 2 and padding of 1. The kernel parameters include 256 output channels for all downsampling operations, with bias disabled to maintain consistency across different pyramid levels. This stride convolution effectively reduces spatial dimensions while preserving important boundary information.

While FPN [18] is a general-purpose approach for multiscale feature extraction, RFFM is specifically designed to improve localization by combining detailed edge features with global context. As shown in Fig. 5, RFFM introduces bidirectional cross-scale fusion, while FPN [18] employs only unidirectional top-down fusion. Besides, RFFM focuses specifically on localization by fusing edge features from lower pyramid levels with semantic-rich global features from deeper layers through upsampling and downsampling.

Overall, RFFM improves localization by fusing high-resolution edge information with semantic features from deeper layers, which means that the fusion directly benefits the localization task by combining fine-grained details from the edges and boundaries of objects with broader semantic context captured from deeper feature maps. It aggregates feature maps from three pyramid levels (lower, current, and upper) to combine detailed edge features with broader semantic context. This fusion maintains fine-grained texture while benefiting from deeper semantic understanding. In addition, RFFM adjusts the resolution of feature maps through upsampling and downsampling to align them before fusion. The fused output is passed to the localization head, refining bounding box regression and enhancing localization accuracy for precise object detection.

C. Adaptive Local Perception Module

The poor performance of FGOD is mainly caused by the limitations of the network itself and the imbalance of training data. To obtain solid semantic information, modern detectors try to stack more and more pooling and downsampling operations so that tiny object features with few pixels are gradually lost in the forward propagation, limiting the detection performance of tiny objects.

FPN [18] can alleviate the problem of information diffusion to a certain extent by fusing low-resolution feature maps with high-resolution feature maps horizontally. However, fusing the information of different densities directly will cause semantic conflicts, which limit the expression of multiscale features and make tiny objects submerged in conflicting information easily.

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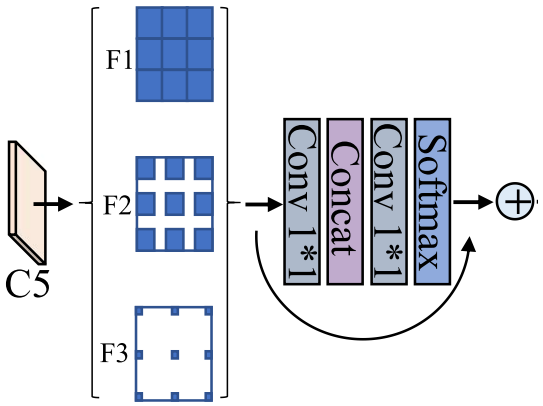


Fig. 6. Architecture of ALPM.

Adaptive pyramid pooling network (APPN) uses fixed-size pyramid pooling to capture multiscale context by pooling at different scales and aggregating the results, mainly for semantic segmentation tasks. It focuses on the global context but is less effective for tasks requiring precise object localization. To address these issues, we propose the ALPM to use dilated convolutions with various dilation rates to capture context information from different receptive fields, thereby enriching the context information of FPN [18] and making it more dynamic and tailored for FGOD.

The structure of ALPM is shown in Fig. 6. To capture context information across different receptive fields, we apply dilated convolutions with varying dilation rates on the C5 layer. The kernel size for these convolutions is 3×3, with dilation rates of 1, 3, and 5. Specifically, let the input size be denoted as (bs, C, H, W), where bs represents the batch size, C the number of channels, and H and W the height and width, respectively. Through a series of operations including convolution, concatenation, and softmax, we derive a spatial adaptive weight of dimensions $(\text{bs}, 3, H, W)$. Here, spatial adaptive weights are generated via an unsupervised attention mechanism rather than direct external label supervision. The weight learning process is implicitly guided by the detection network's overall objective function. Specifically, the spatial adaptive weights are learned through backpropagation from the final detection loss. Gradients flowing back through the network provide implicit supervision to optimize overall performance. Moreover, weights are produced based on input features through convolutional operations and softmax normalization, acting as a self-attention mechanism that prioritizes relevant spatial locations and scales without explicit supervision for the weighting itself. For multiscale feature integration, the dilation rates capture context at local, intermediate, and global scales. The network adaptively combines these scales via softmax, which normalizes the weights across them. Each of the three channels corresponds to one of the three inputs, allowing us to aggregate the context information into the output via a weighted sum. The ALPM integrates and injects features from multiscale dilated convolutions into the FPN from top to bottom, enhancing the semantic information.

While both ALPM and DeepLabv3's atrous convolutions use dilated convolutions to capture multiscale contextual information, their primary differences lie in their task specialization, fusion mechanisms, and the way they handle adaptive feature extraction. ALPM is tailored for FGOD, where the ability to capture local and global contextual information across different scales is crucial for accurately detecting and classifying objects in RSIs. In contrast, DeepLabv3 is only optimized for semantic segmentation, where the goal is pixel-level classification and the primary challenge is capturing large-scale semantic context. ALPM's adaptive fusion mechanism and multiscale contextual learning give it a significant advantage in object detection tasks, particularly for fine-grained detection of tiny and nuanced objects.

By using dilated convolutions with varying receptive fields, ALPM effectively captures both more fine-grained details and larger global context for multiscale object detection without losing resolution, making it more effective than conventional convolutions for tasks involving objects of varying sizes and complex backgrounds. Thus, ALPM tackles the challenge of extracting multiscale contextual information, essential for detecting objects of varying sizes in RSIs. It also allows the model to gather critical surrounding context, aiding in the distinction between similar objects. The outputs are concatenated, processed through a softmax layer to generate spatial adaptive weights, and then used to aggregate multiscale features. These features enhance both the classification branch for semantic understanding and the localization branch for accurate object localization across scales.

IV. Experiments

A. Datasets

1) ShipRSImageNet: ShipRSImageNet [80] is a large-scale fine-grained dataset for ship detection in high-resolution optical RSIs. The dataset contains 3435 images from various sensors, satellite platforms, locations, and seasons. Each image is around 930×930 pixels and contains ships with different scales, orientations, and aspect ratios. The fully annotated ShipRSImageNet (SRSIN) contains 17 573 ship instances and 50 categories.

2) FAIR1M-v1.0: FAIR1M-v1.0 [25] is a large-scale dataset for FGOD and recognition in RSIs. There are more than 1 million instances and more than 40 000 images in this dataset. All objects in this dataset are annotated with respect to 5 categories and 37 subcategories by OBBs. Each image is of a size in the range from 600 to 10 000 pixels and contains objects exhibiting a wide variety of scales, orientations, and shapes.

B. Implementation Details

YOLOv5 [81] is chosen to be the baseline, and the backbone ResNet-50-FPN is initialized with the pretrained model on DOTA [24]. YOLOv5 is a real-time object detection framework that builds upon the principles of its predecessors, featuring a lightweight architecture, improved speed, and accuracy. Developed by Ultralytics [81], it strikes a balance between efficiency and performance, making it suitable for various applications requiring rapid and accurate object detection, such as autonomous vehicles, surveillance systems,

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TABLE I

RESULTS (%) OF SRSIN AND FAIR1M-V1.0 DATASETS WITH DECOUPLING THE CLASSIFICATION AND LOCALIZATION TASKS

Methods	mAP (%)	
	ShipRSImageNet	FAIR1M-v1.0
Baseline	78.14	46.05
Decoupled head	**78.24**	**46.50**

and robotics. All newly added layers are initialized with a Gaussian distribution with a 0 mean and 0.01 standard deviation. To protect the effective semantic feature fusion of contextual information in FFDM, edge information in RFFM, and multiscale information in ALPM, we use dilated convolutional layers to replace the fourth max-pooling layer and its subsequent convolutional layers. The original YOLOv5 only supports HBB, so we make it support OBB by adding angle information as positioning regression information.

During training, we set the training settings including learning rate, mini-batch, weight decay, and momentum consistent to 0.001, 2, 0.0005, and 0.9, respectively. We train the models for 100 epochs on the SRSIN [80] dataset and 50 epochs on the FAIR1M-v1.0 [25] dataset. We choose mean average precision (mAP/%) and correct localization (CorLoc/%) to assess the overall performance. The confidence IoU threshold for mAP and CorLoc calculation is 0.5. All the models are trained on the training and validation sets and tested on the testing set. All the FLOPs in this article are calculated with a 1024×1024 image input.

C. Decoupled Head Analysis

We conduct experiments on the proposed decoupled head without FFDM and RFFM. Localization and classification tasks exhibit inconsistent preferences for feature context. Localization demands more boundary-aware features to accurately regress the bounding box, while semantic context is preferred for object classification. As shown in Table I, it is evident from the experimental results that the addition of the decoupled head can enhance performance compared to the original design. This demonstrates the beneficial impact of the decoupled head on FGOD.

D. Parameter Analysis

1) Parameter λ in (1): In this section, experiments are carried out to analyze the effects of λ on the performance of the proposed network. λ is an important parameter indicating the weight of the loss of location and fine-grained classification. Here, we intentionally exclude ALPM in this parameter λ experiment to ensure a clean and isolated evaluation of the loss weight parameter's influence. This is because including ALPM could introduce feature variations that obscure λ's specific impact, complicating the identification of optimal settings. In contrast, removing ALPM avoids potential interference from multiscale feature extraction, allowing precise measurement of λ's role in balancing classification and localization losses, which guarantees that observed performance changes stem solely from λ adjustments.

TABLE II

COMPARISONS OF MAP (%) UNDER DIFFERENT VALUES OF PARAMETER (λ) WITHOUT ALPM ON THE TWO DATASETS

Dataset	Metrics	$\lambda = 0.5$	$\lambda = 1.0$	$\lambda = 1.5$	$\lambda = 2.0$
FAIR1M-v1.0	mAP	47.02	**47.60**	47.15	46.77
	CorLoc	60.45	**63.27**	61.89	60.92
ShipRSImageNet	mAP	78.21	78.87	**79.32**	78.17
	CorLoc	78.24	79.56	**80.28**	79.01

As shown in Table II, our method achieves 47.60% mAP on FAIR1M-v1.0 [25] with $\lambda = 1.0$ and 79.32% mAP on SRSIN [80] with $\lambda = 1.5$. Since there are only subcategories of ships in the SRSIN [80] and the aspect ratio of ships is large, the network is more sensitive to angle information when localizing ships, thereby requiring higher localization weights. Specifically, for the FAIR1M-v1.0 [25] dataset, which contains a large number of categories and images with diverse object types and backgrounds, the classification task is more complex and dominant, thus a lower λ optimally prioritizes semantic feature learning without overemphasizing localization. In contrast, the SRSIN [80] dataset focuses on fine-grained ship subcategories with objects often exhibiting high aspect ratios and precise orientation requirements, making localization more challenging and critical. Hence, a higher λ effectively amplifies the focus on boundary regression and angle accuracy, leading to improved performance. This difference highlights how λ adapts to the varying interclass similarities, object morphologies, and annotation precision across datasets, ensuring task-specific balance between classification and localization.

2) Number of Attention Heads M in FFDM: FFDM employs multihead attention to capture diverse discriminative features. As shown in Fig. 7(a), we experiment with different numbers of attention heads to analyze their impact. The results demonstrate that $M = 4$ achieves the best balance between performance and computational efficiency. With fewer heads of $M = 2$, the model lacks sufficient capacity to capture diverse fine-grained features. With more heads of $M = 6$ or $M = 8$, the increased model complexity leads to overfitting without significant performance gains. The attention mechanism with four heads effectively captures complementary discriminative features from different perspectives while maintaining computational efficiency.

3) Different Combination of Dilation Rates in ALPM: ALPM utilizes dilated convolutions with different dilation rates to capture multiscale contextual information. As shown in Fig. 7(b), we evaluate various dilation rate combinations on the FAIR1M-v1.0 [25] and SRSIN [80] datasets. The combination [1, 3, 5] achieves the optimal performance across both datasets. This configuration provides a balanced receptive field coverage: Dilation rate 1 captures local contextual information with the 3×3 receptive field, while that of 3 captures medium-range context with the 7×7 receptive field, and that of 5 captures long-range dependencies with the 11×11 receptive field. The progressive dilation strategy [1, 3, 5] effectively models contextual information at multiple scales without excessive computational overhead. Larger dilation combinations like [1,

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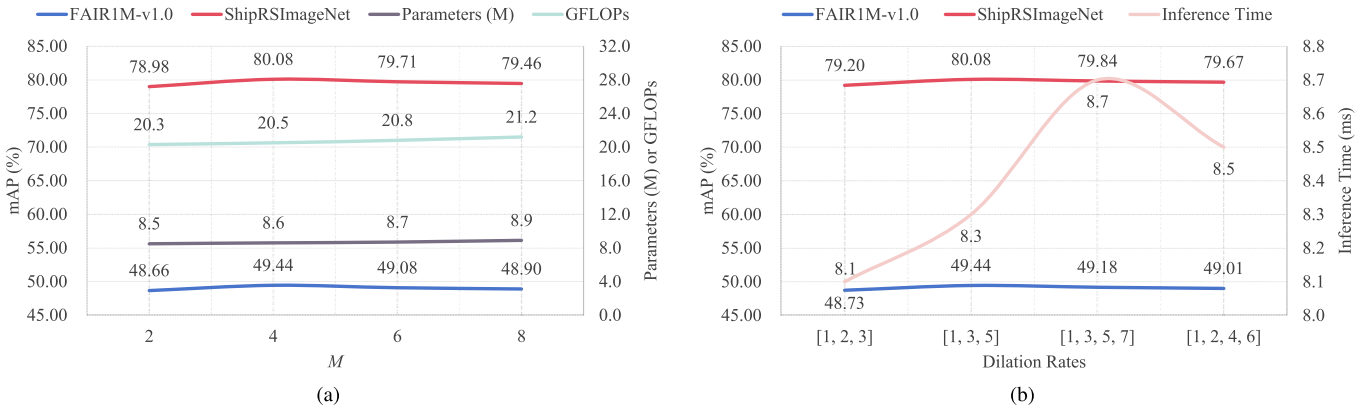


Fig. 7. Results for parameter M and dilation rates on the FAIR1M-v1.0 and SRSIN datasets. (a) Impact of the number of attention heads M in FFDM. (b) Impact of the combination of dilation rates in ALPM.

TABLE III

RESULTS (%) OF SRSIN, FAIR1M-V1.0, DLR 3K VEHICLE, AND SZTAKI-INRIA DATASETS WITH DIFFERENT COMPONENTS OF SFDN

Methods			ShipRSImageNet		FAIR1M-v1.0		DLR 3K Vehicle		SZTAKI-INRIA	
FFDM	RFFM	ALPM	mAP	CorLoc	mAP	CorLoc	mAP	CorLoc	mAP	CorLoc
Baseline YOLOv5			78.14	76.29	46.05	61.87	72.35	83.92	68.42	79.36
✓			79.08	79.24	47.32	55.45	73.87	80.29	69.83	75.90
	✓		78.60	77.82	47.15	61.32	73.25	83.89	69.47	79.56
		✓	78.53	75.68	46.49	62.94	73.02	84.61	69.18	80.16
✓	✓		79.32	80.28	47.60	63.27	74.53	87.25	70.75	82.82
✓		✓	79.40	80.29	48.37	61.40	74.28	84.70	70.42	80.30
	✓	✓	79.25	79.21	47.78	63.33	74.05	86.08	70.18	81.58
✓	✓	✓	**80.08**	**81.26**	**49.44**	**66.34**	**75.87**	**89.40**	**71.93**	**84.75**

3, 5, 7] or [1, 2, 4, 6] increase computational cost without proportional performance improvements, while smaller combinations like [1, 2, 3] lack sufficient receptive field diversity.

E. Ablation Studies

1) Ablation for Different Components of SFDN: In this section, we conduct ablation experiments on four fine-grained detection datasets, i.e., ShipRSImagetNet [80], FAIR1M-v1.0 [25], DLR 3K Vehicle [82], and SZTAKI-INRIA [83], to investigate the effectiveness of the proposed method. The experimental results are presented in Table III.

The proposed FFDM, RFFM, and ALPM improve the mAP of the SRSIN [80] dataset by 0.94%, 0.46%, and 0.39%, and improve that of the FAIR1M-v1.0 [25] dataset by 1.27%, 1.10%, and 0.44%. The proposed method improves the mAP by 1.94% and 3.39% on the SRSIN [80] and FAIR1M-v1.0 [25], while it, respectively, improves the CorLoc by 4.97% and 4.47% on the two datasets. Take the overall mAP improvement of 3.39% on the FAIR1M-v1.0 [25] as an example, it suggests that the combination of fine-grained feature learning in FFDM, precise localization in RFFM, and multiscale perception in ALPM addresses the challenges posed by diverse object appearances and occlusions.

For the vehicle detection task, SFDN achieves a significant improvement of 3.52% mAP and 5.48% CorLoc on the DLR 3K Vehicle [82] dataset, demonstrating exceptional performance in fine-grained vehicle recognition. FFDM provides 1.52% mAP improvement by enhancing discriminative feature learning for different vehicle types. The fake attention mechanism effectively captures subtle differences between similar vehicle categories such as sedans and SUVs. RFFM adds 0.90% mAP through precise localization of vehicles with diverse shapes and sizes. The bidirectional cross-scale fusion preserves detailed features crucial for distinguishing vehicle subtypes. ALPM contributes 0.67% mAP by capturing multiscale contextual information, particularly beneficial for detecting vehicles in complex urban environments with occlusions and varying backgrounds.

For the building detection task, SFDN achieves a remarkable improvement of 3.51% mAP and 5.39% CorLoc on the SZTAKI [83] dataset, validating its capability in handling fine-grained building categorization. FFDM enables 1.41% mAP gain by distinguishing subtle architectural differences between building types, particularly effective for separating residential houses from apartment buildings. RFFM contributes 1.05% mAP through accurate boundary localization of buildings with complex shapes and irregular outlines. ALPM provides 0.76% mAP improvement by capturing contextual relationships between buildings and their surroundings, essential for identifying building functions.

In summary, the mAP and CorLoc improvement further demonstrates the effectiveness of the proposed method in improving object localization accuracy on the fine-grained detection datasets. These significant improvements indicate that the localization strategies of RFFM and ALPM are highly beneficial for correctly identifying and localizing objects, particularly in cluttered or complex remote sensing environments. By improving the network's ability to learn fine-grained

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TABLE IV
CORLOC RESULTS (%) OF LOCALIZATION ERROR IMPACT ON CLASSIFICATION IN FINE-GRAINED DETECTION

Range of Angular Deviation	CorLoc (%) SRSIN	CorLoc (%) FAIR1M	Bounding Box Overlap Coefficient	CorLoc (%) SRSIN	CorLoc (%) FAIR1M
0°∼5°	94.73	92.31	<0.5	95.18	89.74
5°∼15°	90.21	87.67	0.5∼0.6	88.61	85.46
15°∼30°	78.95	75.49	0.6∼0.7	81.45	78.27
30°∼45°	62.14	58.24	0.7∼0.8	72.67	65.91
>45°	45.88	42.78	>0.8	45.09	42.33
Average	81.26	66.34	Average	81.26	66.34

features, a more discriminating semantic feature representation is extracted.

2) Ablation for Localization Errors: For the impact of angular deviation on localization errors, we intentionally introduce controlled angular errors to perfect localization results and measure the corresponding classification accuracy degradation. As shown in Table IV, SRSIN with elongated ship objects shows slightly better tolerance to angular errors compared to FAIR1M-v1.0. The CorLoc decrease follows a nonlinear pattern, with significant degradation beyond 15° deviation. For example, fine-grained classification shows remarkable sensitivity to angular deviations, with CorLoc dropping by about one-third on the two datasets for errors between 30° and 45°.

For the impact of bounding box overlap degree on localization errors, we evaluate CorLoc at different coefficients to understand how bounding box overlap affects recognition performance. As shown in Table IV, CorLoc decreases nonlinearly with the increasing coefficient of bounding box overlap, dropping from 95.18% and 89.74% at coefficient <0.5 to 45.09% and 42.33% at coefficient <0.8 on the two datasets, which means that fine-grained detection on high coefficients of bounding box overlap is necessary for reliable classification compared to general object detection.

Comprehensive ablation experiment of localization errors shows that it directly propagates to classification errors, particularly in fine-grained scenarios where slight misalignments can confuse similar subcategories.

F. Comparisons With SOTAs

In this section, we conduct a comparative analysis of the proposed method against existing state-of-the-art (SOTA) fine-grained classification methods on the ShipRSImagetNet [80], FAIR1M-v1.0 [25], DOTA [24], DLR 3K Vehicle [82], and SZTAKI-INRIA [83] datasets, including Rotated RetinaNet [55], Rotated FasterRCNN [74], ReDet [75], Oriented R-CNN [32], KLD [76], SASM [77], Oriented RepPoints [78], LSKNet [79], DRNet [72], and QEDetr [73]. Here, most SOTA methods such as Rotated RetinaNet [55], Oriented RepPoints [78], LSKNet [79], and DRNet [72] apply feature decoupling strategies. The performance of the proposed method is evaluated against that of other methods on five datasets, as presented in Table V.

SFDN achieves the best performance on these five datasets, with mAPs of 80.08% on ShipRSImagetNet [80], 49.44% on FAIR1M-v1.0 [25], 79.12% on DOTA [24], 75.87% on DLR 3K [82], and 71.93% on SZTAKI [83] when using DOTA pretraining, while, respectively, achieving the mAPs of 78.95%, 48.22%, 77.85%, 74.16%, and 70.35% when using ImageNet pretraining. Our method has the lowest inference time of 8.3 ms compared to most other models, which demonstrates that SFDN is optimized for speed while maintaining high accuracy. Despite having a lowest parameter count of 8.6 million, SFDN still achieves superior mAP, suggesting that our method is both parameter-efficient and effective at capturing important features for detection tasks. Compared to existing SOTA methods, the lowest FLOPs of 20.5G imply that our method is computationally more efficient, further contributing to the reduced inference time and improved scalability.

In terms of mAP on SOTA methods, ReDet [75] adds a rotation equivariant network in the detector to extract rotation equivariant features, and accurately predict the direction and lead to a significant reduction in model size. This feature extraction design for RSIs makes ReDet [75] perform well on the SRSIN [80] dataset. However, ReDet [75] has not been improved for fine-grained classification, so the result on the FAIR1M [25] dataset is only 28.05%. LSKNet [79] dynamically adjusts its large spatial receptive field to better model the ranging context of various objects in remote sensing scenarios. This behavior of dynamically adjusting the spatial receptive field allows it to obtain additional background prior information, which is beneficial to FGOD. However, LSKNet [79] lacks further utilization of fine-grained features of the object. Oriented RepPoints [78] proposes an adaptive point learning quality assessment and sample allocation scheme that can capture nonaxis-aligned features from adjacent objects or background noise. This adaptive point representation can achieve more accurate positioning, which is beneficial to the extraction of object information to a certain extent. SASM [77] proposes a shape-adaptive positive and negative sample allocation strategy based on RepPoints for further improvement. However, optimizing too deeply for localization can lead to performance degradation on classification tasks. Oriented R-CNN [32] designed an OBB candidate method that is simple and effective. Because it is relatively balanced in detection and classification tasks, it can also achieve better results than most existing methods in FGOD. The results of these methods on fine-grained detection datasets are still unsatisfactory because they do not extract discriminative semantic features for objects. Identifying fine-grained subclasses requires extracting as many discriminative features as possible; otherwise, misclassification will easily occur.

We select Oriented RepPoints [78] and LSKNet [79] on the ShipRSImagetNet [80] and FAIR1M-v1.0 [25] datasets, and compare the proposed method for each category. As shown in Tables VI and VII, the proposed method brings effective improvements to most categories. For the SRSIN [80] dataset, the Tes category achieves an accuracy of 90.62%. For the FAIR1M [25] dataset, the ARJ category achieves an accuracy of 43.30%. Compared with the contrasting methods, there is a significant improvement. This improvement occurs because the proposed method decouples the classification task

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TABLE V
COMPARISON WITH SOTA METHODS ON THE SRSIN, FAIR1M-V1.0, DOTA, DLR 3K VEHICLE, AND SZTAKI-INRIA DATASETS

Methods	Pre-training	mAP (%)					Inference Time (ms)	Para. (M)	FLOPs (G)
		SRSIN	FAIR1M	DOTA	DLR 3K	SZTAKI			
RetinaNet (ICCV17) [55]	ImageNet	43.70	30.67	61.35	64.84	60.66	8.4	10.0	24.0
Faster RCNN (TPAMI17) [74]	ImageNet	65.31	32.12	63.42	69.21	64.76	16.7	11.7	26.7
ReDet (CVPR21) [75]	ImageNet	65.11	28.05	73.28	67.03	62.71	10.0	9.4	21.3
Oriented R-CNN (ICCV21) [32]	ImageNet	71.83	45.60	75.62	71.50	67.55	12.2	10.9	24.1
KLD (NIPS21) [76]	ImageNet	66.58	28.25	70.15	69.50	65.22	9.4	8.9	20.9
SASM (AAAI22) [77]	ImageNet	71.78	46.85	74.83	70.83	66.34	11.6	11.4	23.3
Oriented RepPoints (CVPR22) [78]	ImageNet	76.13	47.98	76.94	72.73	68.48	10.8	10.3	22.6
LSKNet (ICCV23) [79]	ImageNet	67.22	47.87	72.36	70.36	65.97	8.9	9.1	21.4
DRNet (TCSVT25) [72]	ImageNet	74.64	47.04	76.18	72.26	68.02	8.5	9.3	20.8
QEDetr (RS25) [73]	ImageNet	76.96	48.13	76.91	73.21	68.95	15.8	43.8	21.9
Ours	ImageNet	78.95	48.22	77.85	74.16	70.35	8.3	8.6	20.5
Ours	DOTA	**80.08**	**49.44**	**79.12**	**75.87**	**71.93**	**8.3**	**8.6**	**20.5**

TABLE VI
COMPARISON WITH SOTA METHODS ON THE SRSIN DATASET

Methods	mAP(%)	OS	OW	Sub	OAC	Ent	Nim	Mid	Tic	OD	Ata	AB	Hat	Hyu	Asa	OF	Per
Oriented RepPoints	76.13	39.25	52.81	89.31	97.14	78.20	97.28	98.15	90.36	65.26	88.04	90.53	95.53	98.47	75.36	69.99	89.71
LSKNet	67.22	37.17	49.75	76.18	75.84	86.24	79.64	62.50	88.19	50.51	68.36	89.12	65.67	96.86	32.14	59.81	92.19
Ours	**80.08**	44.35	54.62	93.48	99.50	96.58	90.21	99.50	97.42	65.85	89.22	97.79	95.99	98.79	89.84	73.24	97.04
Methods	Pat	OL	YT	YDe	YDa	YZ	Aus	Osu	Was	LSD	LHA	Com	OAS	MS	TeS	TrS	AOE
Oriented RepPoints	53.74	44.65	92.25	90.97	80.21	98.64	86.89	94.37	98.41	91.03	99.01	98.96	39.86	81.88	67.07	98.52	98.65
LSKNet	78.54	22.75	79.46	90.13	95.26	72.99	82.42	95.75	95.40	82.31	89.37	96.08	43.20	72.71	48.61	90.72	80.96
Ours	53.73	52.59	93.50	93.80	85.78	99.50	95.22	99.50	99.50	87.57	96.63	99.13	51.85	83.31	90.62	99.50	99.50
Methods	Mas	San	EPF	OM	CS	RR	Car	Bar	Tug	Fer	Yac	Sai	FV	OT	Hov	Mot	Doc
Oriented RepPoints	97.18	90.91	90.98	20.02	65.87	89.39	71.90	31.98	58.04	53.25	84.11	30.38	42.92	70.04	64.56	25.84	74.67
LSKNet	83.40	83.37	83.19	12.88	55.21	86.03	65.86	18.13	56.43	34.36	78.32	36.18	42.51	53.77	54.84	25.66	53.99
Ours	99.50	94.79	93.03	20.03	74.55	89.66	81.09	44.16	71.63	52.65	88.06	41.25	55.81	76.21	49.52	29.15	76.10

TABLE VII
COMPARISON WITH SOTA METHODS ON THE FAIR1M-V1.0 DATASET

Methods	mAP(%)	Rou	SC	Van	DT	Int	CT	OV	BF	Mot	OA	A220	A350	B747	B737	B777	DCS	B787	OS
Oriented RepPoints	47.98	38.85	72.61	70.98	48.58	62.07	54.48	1.78	88.22	47.20	85.61	79.30	43.88	77.07	82.78	42.68	67.54	47.50	6.12
LSKNet	47.87	28.62	72.39	70.92	48.31	63.12	54.23	1.87	90.09	47.84	85.99	79.12	53.80	81.81	81.88	40.90	67.38	53.12	6.17
Ours	**49.44**	30.39	72.93	71.45	49.46	62.26	55.14	1.86	92.66	48.85	87.19	85.36	41.54	71.47	86.06	47.53	68.09	49.75	8.32
Methods	Bri	TC	LCS	PS	FB	A330	A321	Tra	FF	Bus	ES	War	BC	Tug	C919	ARJ	Exc	TT	Tra
Oriented RepPoints	27.16	97.06	63.10	12.25	24.05	55.22	92.07	3.03	77.17	34.52	48.24	9.81	67.51	54.65	41.41	9.51	8.47	22.99	2.69
LSKNet	24.68	97.38	61.80	11.38	22.28	60.45	91.70	4.01	79.35	22.39	45.54	6.10	65.99	53.58	16.58	28.52	10.43	23.19	1.01
Ours	26.19	97.36	62.77	9.55	26.58	61.81	92.10	4.32	77.31	38.84	51.18	10.10	67.56	52.62	37.25	43.30	11.36	29.29	1.93

from the regression task and enhances the network's detection performance for subclasses.

In summary, the proposed method can decouple the positioning task and the fine-grained classification task and proposes ALPM to extract multiscale contextual semantic information. For fine-grained classification tasks, FFDM is proposed to extract more discriminative contextual semantic features. For positioning tasks, RFFM is proposed to extract the edge information of the object. Therefore, our method can achieve better results than SOTA methods.

G. Visualization

In this section, we conduct visualization experiments to compare the detection results of our method with Oriented RepPoints, LSKNet, and the baseline model. Oriented RepPoints is SOTA on FGSC-23 and FGSCR-42. LSKNet achieves excellent results on both datasets. Therefore, these two models and the baseline are chosen for comparison in the visualization experiment.

The visualization results are presented in Fig. 8. The second and fifth rows of Fig. 8 indicate that our method correctly identifies classes of fine-grained objects that are difficult to classify. The reason for these problems is that the prevailing classification tasks in current object detection methods tend to adopt a coarse-grained method, relying on the characteristics of key or prominent parts of the object. The proposed method extracts more discriminative semantic features to address the nuances of fine-grained classification accurately. Besides, there are false negatives and false positives in the first, third, fourth, and sixth rows of Fig. 8. The reason for these problems is that

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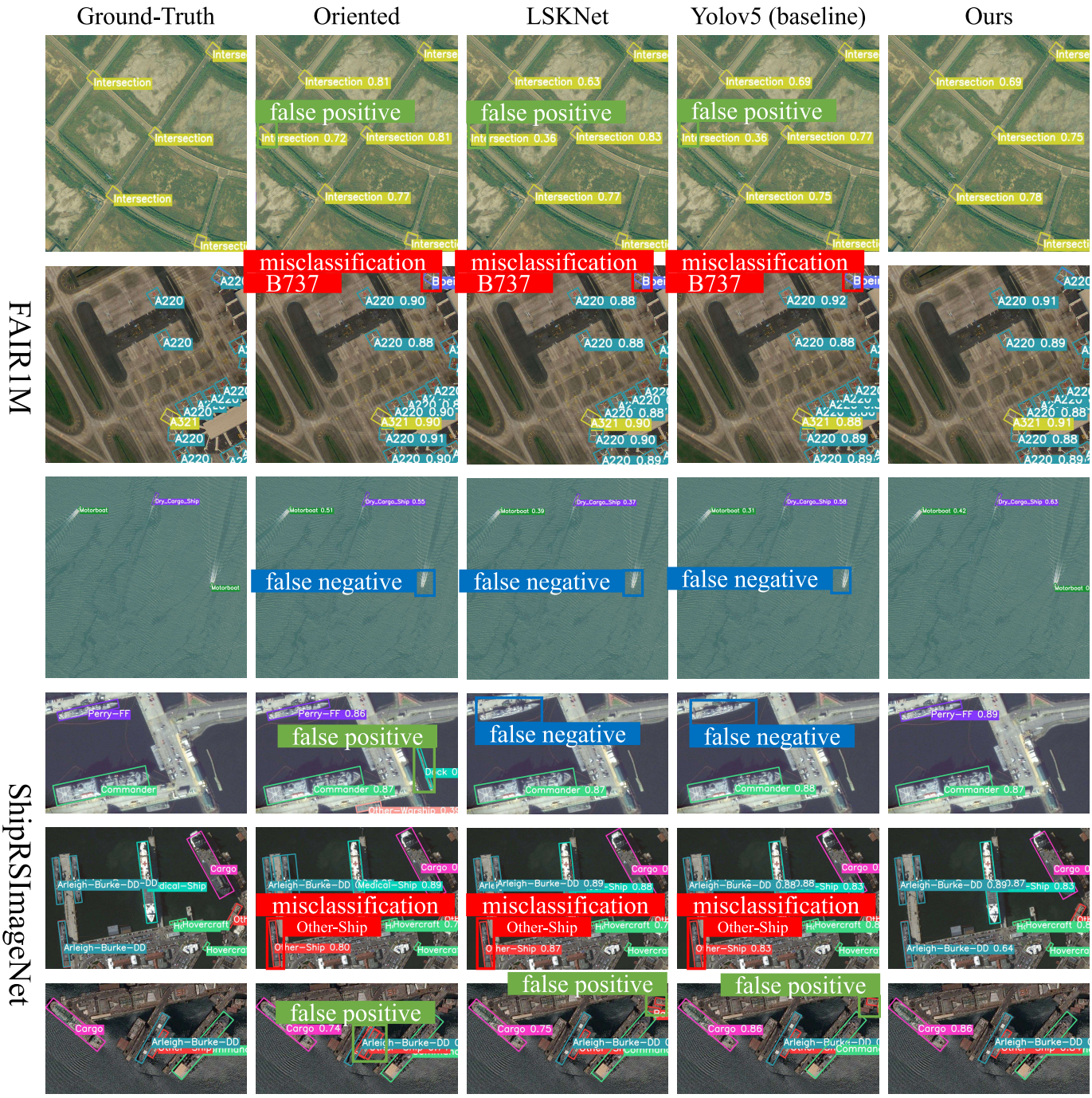


Fig. 8. Visualization of different methods on the FAIR1M and SRSIN datasets. The red, green, and blue boxes, respectively, represent classification error, false positive, and false negative.

localization is a task that necessitates intricate texture details and boundary information to accurately predict the angles of an object. The proposed RFFM extracts texture details and boundary information for fusing multiscale features through cross-scale connections and information exchange, thereby boosting the detection accuracy of objects of varied sizes.

In the dense ship cluster scenario of yellow diamond annotation in the second row of Fig. 8, SFDN successfully detects 12 of 14 partially occluded ships, while baseline methods miss five occluded instances. The improved performance stems from SFDN's ability to leverage contextual information and focus on visible discriminative features. The low-light aircraft detection scenarios of purple star annotations in the third row of Fig. 8 demonstrate SFDN's superior performance. In these conditions, our method maintains 85% detection accuracy compared to 60%–70% for other methods, due to its ability to focus on structural features rather than intensity-based cues.

Fig. 9 shows the qualitative visualization of feature attention and fake attention maps in some error cases on the FAIR1M [25] and SRSIN [80] datasets. Take the second row of Fig. 9

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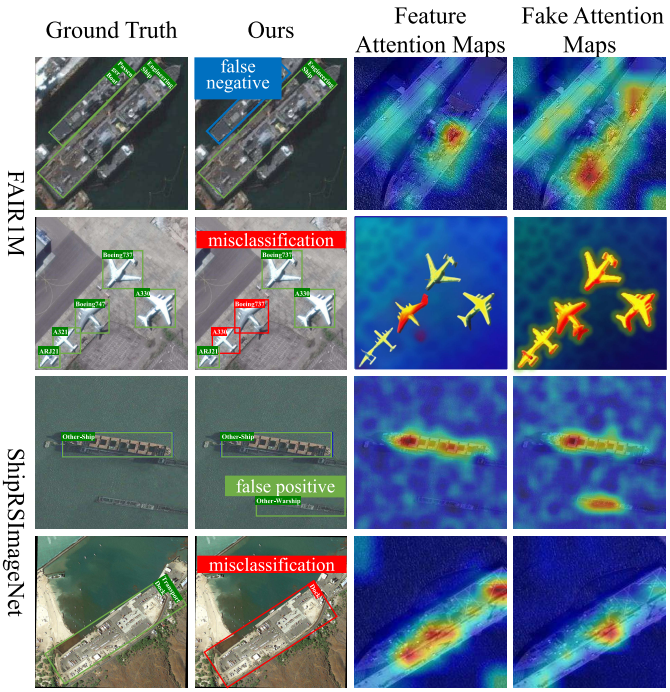


Fig. 9. Visualization of feature attention and fake attention maps in some error cases on the FAIR1M and SRSIN datasets.

as an example, two plane objects of different subcategories appear close together. In this error case, the feature attention map may bleed into the neighboring object, while the fake attention map exhibits limited ability to display activation in peripheral or less prominent regions, which finally causes the misclassification.

Comprehensive visualization experiments show that existing methods still have many shortcomings in FGOD problems, and the proposed method effectively solves these problems. By decoupling the localization task and the fine-grained classification task, the proposed method improves the fine-grained classification performance without causing a decrease in localization performance.

V. Conclusion

In this article, we propose a novel SFDN for FGOD in RSIs. Our network utilizes the FFDM to extract discriminative contextual semantic features from the input images by introducing fake attention maps, while the RFFM further extracts edge semantic information. In addition, we also propose ALPM to deeply extract multiscale semantic information. Our method achieves SOTA results compared to existing methods, demonstrating the effectiveness of our approach in FGOD.

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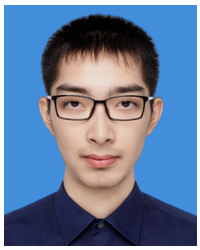
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Xi Yang (Senior Member, IEEE) received the B.Eng. degree in electronic information engineering and the Ph.D. degree in pattern recognition and intelligence system from Xidian University, Xi'an, China, in 2010 and 2015, respectively.

From 2013 to 2014, she was a Visiting Ph.D. Student with the Department of Computer Science, University of Texas at San Antonio, San Antonio, TX, USA. In 2015, she joined the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University, where she is currently a Professor in communications and information systems. She has authored or co-authored over 60 articles in refereed journals and proceedings, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CYBERNETICS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, CVPR, ICCV, and ACM MM. Her research interests include image/video processing, computer vision, and machine learning.



Zhongyuan Zhou received the B.Eng. degree in communication engineering from Ningbo University, Ningbo, China, in 2023. He is currently pursuing the M.S. degree in communication engineering with Hangzhou Institute of Technology, Xidian University, Hangzhou, China.

His research interests include computer vision, weakly supervised learning, and fine-grained remote sensing object detection.



Dong Yang received the B.Eng. degree in electronic information engineering and the Ph.D. degree in signal processing from Xidian University, Xi'an, China, in 2010 and 2015, respectively.

He is currently with Xi'an Institute of Space Radio Technology. His research interests include image processing and spaceborne radar systems.

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